

1 Summary

Model (Run-name): model1 (test)

Iteration: 18

TAC: 154823.1229058172

Took: 1.9434492588043213s

Total Time: 26.9273579120636s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

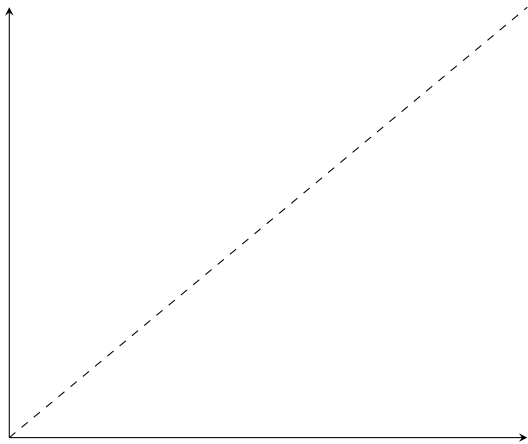
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

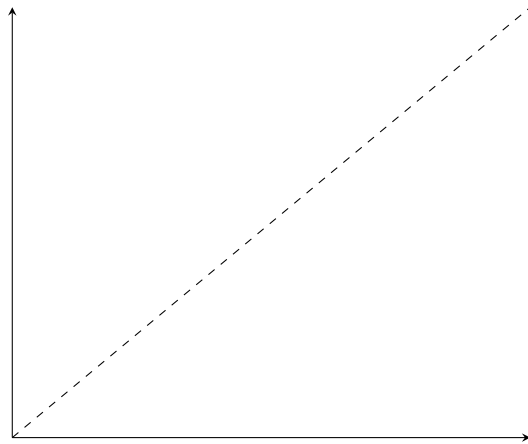
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{1,1,1}$	0.052115641382223776	$b_{1,1,1}^{H,out}$	903.1278250661071
$z_{1,2,2}$	1.0	$RecLMTD_{1,2,2}$	0.017798348027897416	$b_{1,2,2}^{H,out}$	3409.244400339228
$z_{2,1,2}$	1.0	$RecLMTD_{2,1,2}$	0.028801765097091796	$b_{2,1,2}^{H,out}$	5301.534304156642
$z_{cu,1}$	1.0	$RecLMTD_{cu,1}$	0.014637938966270104	$b_{1,1,1}^{C,in}$	5995.308268974677
$z_{cu,2}$	1.0	$RecLMTD_{cu,2}$	0.009571385914187273	$b_{1,2,2}^{C,in}$	4550.0
$z_{hu,1}$	1.0	$RecLMTD_{hu,1}$	0.022526781496037367	$b_{2,1,2}^{C,in}$	4875.935694729515
$A_{1,1,1}^\beta$	70.95870595008942	$q_{1,1,1}$	683.4480838338109	$b_{1,1,1}^{C,out}$	6678.756352808488
$A_{1,2,2}^\beta$	69.41355730879985	$q_{1,2,2}$	1950.0	$b_{1,2,2}^{C,out}$	6500.0
$A_{2,1,2}^\beta$	139.47623233772285	$q_{2,1,2}$	2424.8278742492826	$b_{2,1,2}^{C,out}$	7300.763568978798
$A_{cu,1}^\beta$	4.825601156120121	$q_{cu,1}$	166.55191616618868		
$A_{cu,2}^\beta$	37.78573239590169	$q_{cu,2}$	1975.172125750718		
$A_{hu,1}^\beta$	13.269097009439808	$q_{hu,1}$	491.7240419169066		
$A_{1,1,1}$	70.95870595008942	$t_{1,1}^{(H)}$	650.0		
$A_{1,2,2}$	69.41355730879985	$t_{1,2}^{(H)}$	581.6551916166188		
$A_{2,1,2}$	139.47623233772285	$t_{1,3}^{(H)}$	386.6551916166189		
$A_{cu,1}$	4.825601156120121	$t_{2,1}^{(H)}$	590.0		
$A_{cu,2}$	37.78573239590169	$t_{2,2}^{(H)}$	590.0		
$A_{hu,1}$	13.269097009439808	$t_{2,3}^{(H)}$	468.7586062875359		
$dt_{1,1,1}$	32.78160279446043	$t_{1,1}^{(C)}$	617.2183972055395		
$dt_{1,1,2}$	10.0	$t_{1,2}^{(C)}$	571.6551916166188		
$dt_{1,2,2}$	81.65519161661882	$t_{1,3}^{(C)}$	410.0		
$dt_{1,2,3}$	36.655191616618865	$t_{2,1}^{(C)}$	500.0		
$dt_{2,1,2}$	18.344808383381178	$t_{2,2}^{(C)}$	500.0		
$dt_{2,1,3}$	58.75860628753588	$t_{2,3}^{(C)}$	350.0		
$dt_{cu,1}$	66.65519161661886	$t_{1,1,1}^H$	370.0		
$dt_{cu,2}$	148.75860628753588	$t_{1,2,2}^H$	370.99663669830295		
$dt_{hu,1}$	62.78160279446044	$t_{2,1,2}^H$	392.80942656290057		
$f_{1,1,1}^H$	2.4408860136921815	$t_{1,1,1}^C$	640.8857765174175		
$f_{1,2,2}^H$	9.214174054970886	$t_{1,2,2}^C$	500.0		
$f_{2,1,2}^H$	13.095529115942245	$t_{2,1,2}^C$	613.8951066390871		
$f_{1,1,1}^C$	10.485338007764964	$b_{1,1,1}^{H,in}$	1586.575908899918		
$f_{1,2,2}^C$	13.0	$b_{1,2,2}^{H,in}$	5359.244400339228		
$f_{2,1,2}^C$	11.892526084706134	$b_{2,1,2}^{H,in}$	7726.362178405924		

3 *RecLMTD* Tangent Points

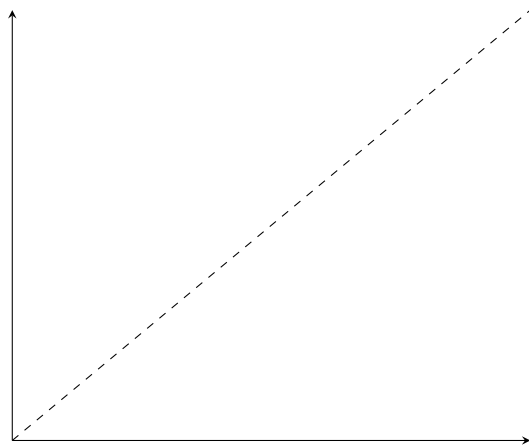
- $(1, 1, 1)$
 - Value: 0.052116
 - Error: 0.000000198988502
 - Relative error: 0.000003818196176
- $(1, 2, 2)$
 - Value: 0.017798
 - Error: 0.000000548171084
 - Relative error: 0.000030798038117
- $(2, 1, 2)$
 - Value: 0.028802
 - Error: 0.000002531457094
 - Relative error: 0.000087884704617
- $cu, 1$
 - Value: 0.014638
 - Error: 0.000000359890776
 - Relative error: 0.000024585560075
- $cu, 2$
 - Value: 0.009571
 - Error: 0.000000008167442
 - Relative error: 0.000000853317929
- $hu, 1$
 - Value: 0.022527
 - Error: 0.000000021247299
 - Relative error: 0.000000943200839



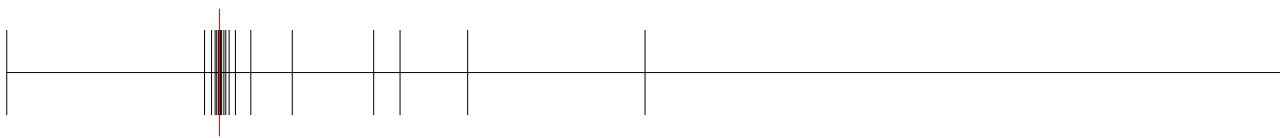
(a) *RecLMTD* - (1, 1, 1)



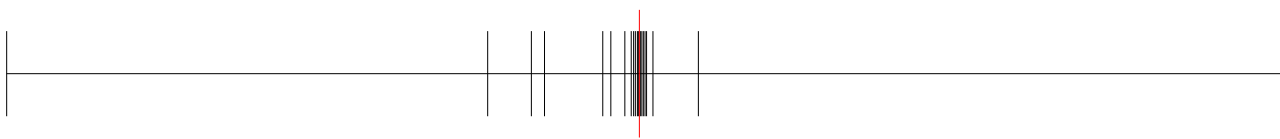
(b) *RecLMTD* - (1, 2, 2)



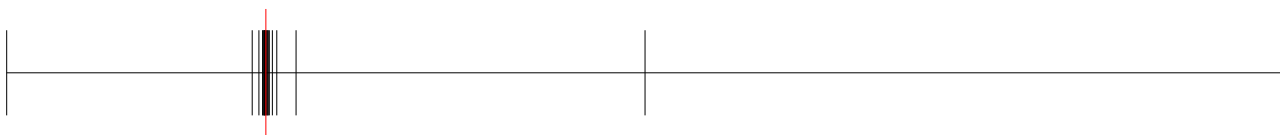
(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - *cu*, 1



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1586.575908899918
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 5359.244400339228
 - Absolute error: 0.22777519373994437
 - Relative error: -4.249955709813783e-05
- (2, 1, 2)
 - Value: 7726.362178405924
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 903.1278250661071
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3409.244400339228
 - Absolute error: 9.18318400773478
 - Relative error: -0.0026863766398869276
- (2, 1, 2)
 - Value: 5301.534304156642
 - Absolute error: 157.48702158560081
 - Relative error: 0.030615391526278388

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 5995.308268974677
 - Absolute error: 1.3103609807803878
 - Relative error: 0.00021861218520494052
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 4875.935694729515
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 6678.756352808488
 - Absolute error: 41.14763834555288
 - Relative error: -0.0061232479511193726
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 7300.763568978798
 - Absolute error: 9.094947017729282e-13
 - Relative error: 1.2457528492463495e-16



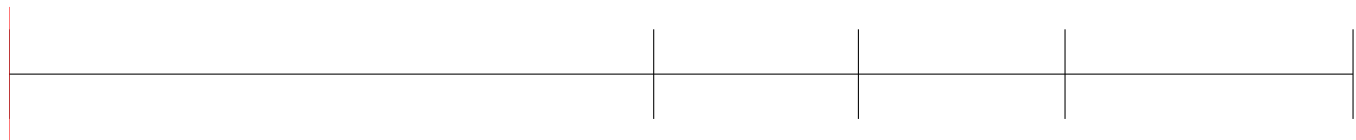
(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



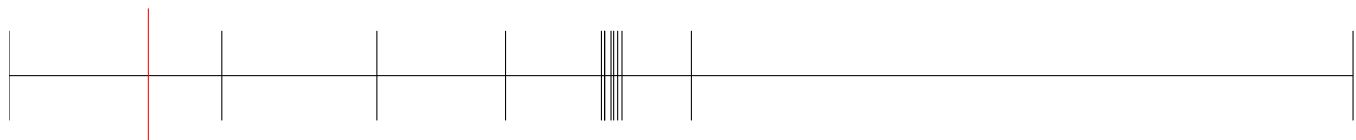
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



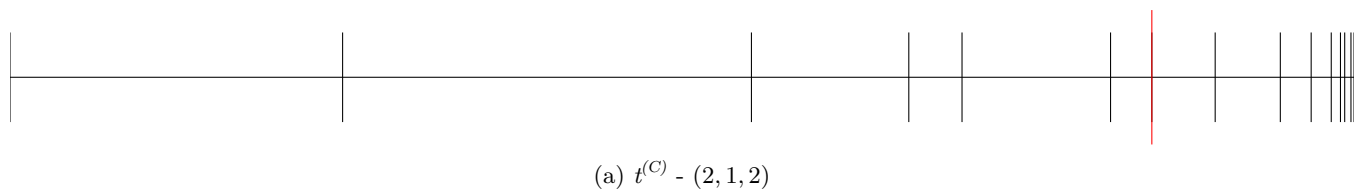
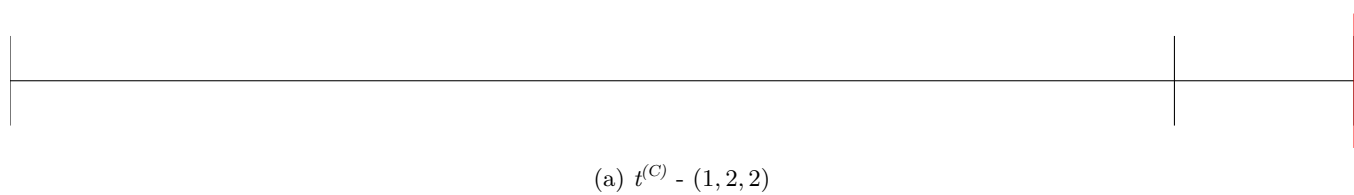
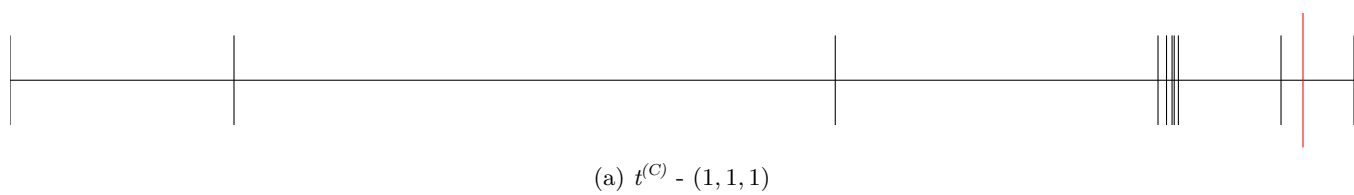
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

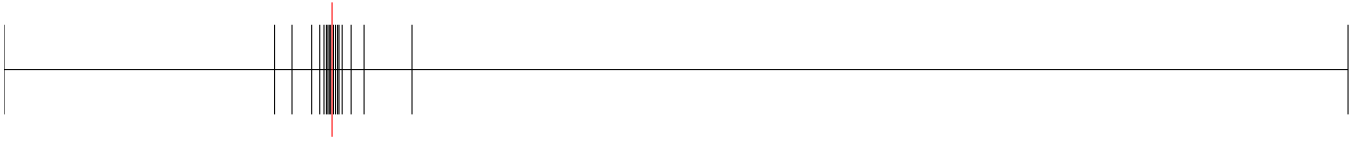
- $(1, 1, 1)$
 - Value: 70.958706
 - Error: 0.277964530812369
 - Relative error: 0.003901986560235
- $(1, 2, 2)$
 - Value: 69.413557
 - Error: 0.0000000000000071
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 139.476232
 - Error: 0.202413332293702
 - Relative error: 0.001449135845517

5.2 A_{CU}

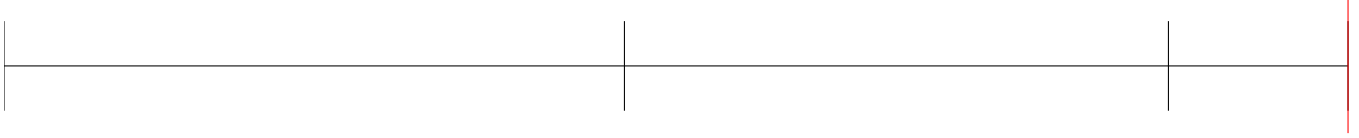
- $cu, 1$
 - Value: 4.825601
 - Error: 0.050352410991889
 - Relative error: 0.010326679755835
- $cu, 2$
 - Value: 37.785732
 - Error: 0.024536929109821
 - Relative error: 0.000648948805387

5.3 A_{HU}

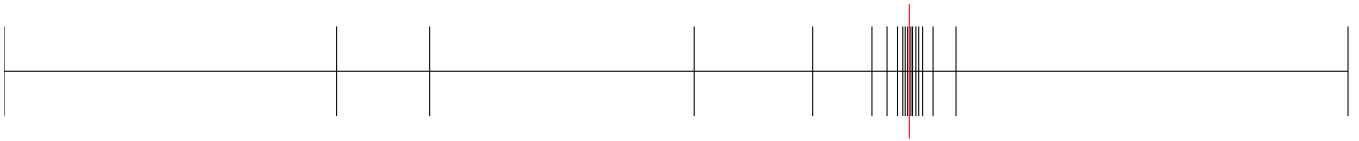
- $hu, 1$
 - Value: 13.269097
 - Error: 0.023255048892761
 - Relative error: 0.001749505940763



(a) $A^\beta - (1, 1, 1)$



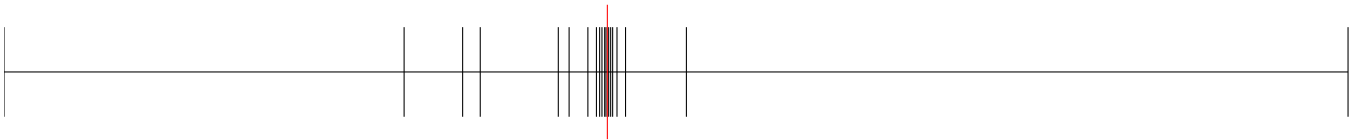
(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



(a) $A^\beta - cu, 1$



(a) $A^\beta - cu, 2$

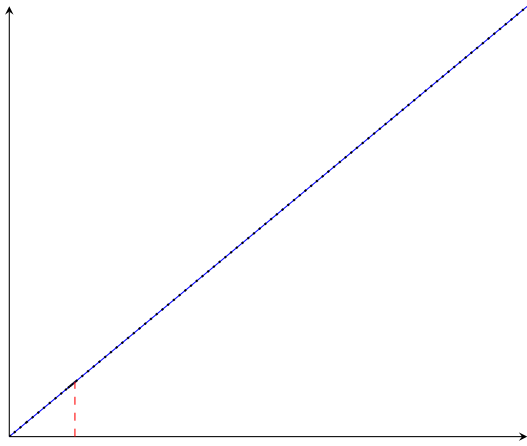


(a) $A^\beta - hu, 1$

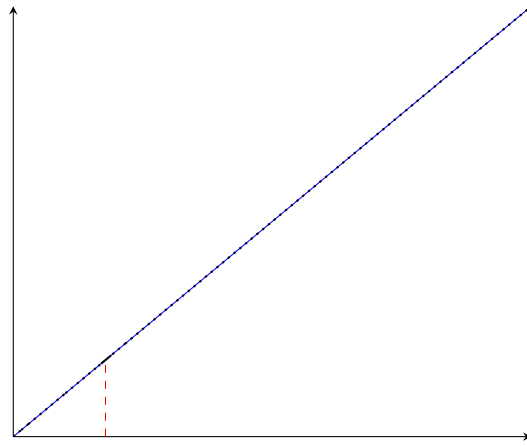
6 A^β Breakpoints

- (1, 1, 1)
 - q : 70.958706
 - $\hat{q}^\beta$: 70.958706
 - q^β : 70.958706
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.22290217272871, 70.30181758862284, 70.50425945122109, **70.60541450524272**, **71.22933662920877**, 71.2616536557244, 71.6637831223933, 71.88123217857378, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1, 2, 2)
 - q : 69.413557
 - $\hat{q}^\beta$: 69.413557
 - q^β : 69.413557
 - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, 67.91344539650363, 68.66676327863097, 69.0314457108091, **69.34502840170047**, **70.37078617087542**, 70.94081287188642, 72.25348351595485, 74.6496383881616, 84.1468297226002, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2, 1, 2)
 - q : 139.476232
 - $\hat{q}^\beta$: 139.476232
 - q^β : 139.476232
 - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 128.52021081477858, 131.69760457629607, 132.12228195956155, **135.81963163113545**, **140.7008934199032**, 142.4338700260237, 144.70843093515643, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 1$
 - q : 4.825601
 - $\hat{q}^\beta$: 4.825601
 - q^β : 4.825601
 - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.269280092716329, 4.432280503220532, **4.613047202566896**, **4.849719301830286**, 4.974201802143407, 4.983938626310074, 4.993068227518925, 5.0418124100090695, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$

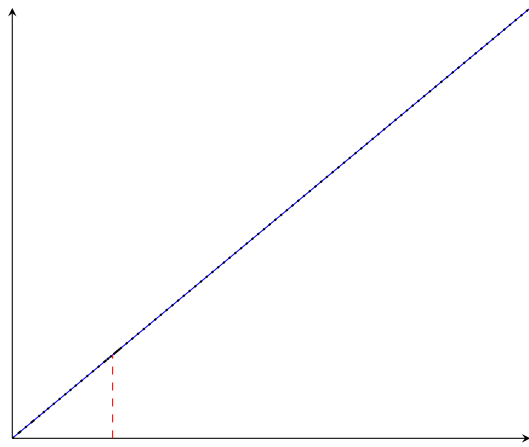
- q : 37.785732
- $\hat{q}^\beta$: 37.785732
- q^β : 37.785732
- Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, 37.554741031910254, 37.6551299822246, **37.682669717644174**, **37.9018730666985**, 37.9454585315849, 38.10872593446926, 38.21652191748154, 38.88483572219945, 285.4001551947793]
- Error: 0.0000000000000000
- Relative error: 0.0000000000000000
- $hu, 1$
 - q : 13.269097
 - $\hat{q}^\beta$: 13.269097
 - q^β : 13.269097
 - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861, 13.161020122653593, 13.192252946941991, 13.213749911872785, 13.218222920629291, 13.22717920580809, **13.244985069124171**, **13.30871858348944**, 13.32663103908035, 13.346967780023082, 13.439500729482463, 13.468285806082731, 14.09178614751222, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



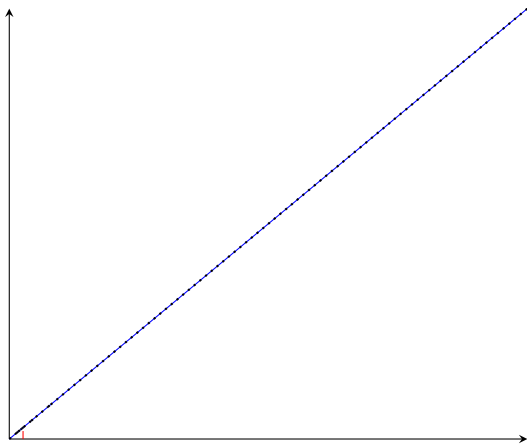
(a) $q^\beta - (1, 1, 1)$



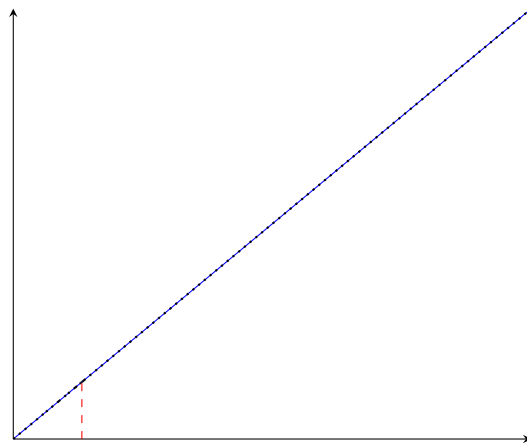
(b) $q^\beta - (1, 2, 2)$



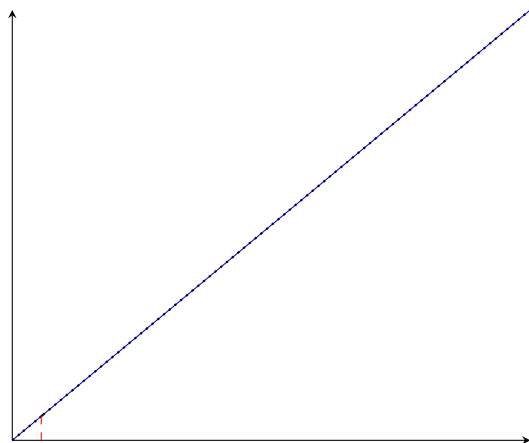
(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 1$



(b) $q^\beta - cu, 2$



(a) $q^\beta - hu, 1$

7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,1}^C$	0.0	$b_{2,2,2}^{H,out}$	4073.637821594076
$z_{1,2,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{1,1,2}^{C,in}$	1274.0643052704852
$z_{2,1,1}$	-0.0	$RecLMTD_{1,1,2}$	0.1	$b_{1,2,1}^{C,in}$	6500.0
$z_{2,2,1}$	0.0	$RecLMTD_{1,2,1}$	0.008082097989248308	$b_{2,1,1}^{C,in}$	2579.5196052746046
$z_{2,2,2}$	-0.0	$RecLMTD_{2,1,1}$	0.029166666666666688	$b_{2,2,1}^{C,in}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{2,2,1}$	0.00608	$b_{2,2,2}^{C,in}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.1	$b_{1,1,2}^{C,out}$	1274.0643052704845
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,2,1}^{C,out}$	6500.0
$A_{2,1,1}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{2,1,1}^{C,out}$	2579.5196052746046
$A_{2,2,1}^\beta$	0.0	$q_{1,2,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,2,2}^\beta$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{hu,2}^\beta$	0.0	$q_{2,2,1}$	0.0		
$A_{1,1,2}$	0.0	$q_{2,2,2}$	0.0		
$A_{1,2,1}$	0.0	$q_{hu,2}$	0.0		
$A_{2,1,1}$	0.0	$t_{1,1,2}^H$	561.2699919745228		
$A_{2,2,1}$	0.0	$t_{1,2,1}^H$	650.0		
$A_{2,2,2}$	0.0	$t_{2,1,1}^H$	590.0		
$A_{hu,2}$	0.0	$t_{2,2,1}^H$	590.0		
$dt_{1,1,3}$	10.0	$t_{2,2,2}^H$	566.261281941302		
$dt_{1,2,1}$	150.0	$t_{1,1,2}^C$	410.0		
$dt_{2,1,1}$	24.166666666666668	$t_{1,2,1}^C$	500.0		
$dt_{2,2,1}$	90.0	$t_{2,1,1}^C$	522.0082244950356		
$dt_{2,2,2}$	220.0	$t_{2,2,1}^C$	350.0		
$dt_{2,2,3}$	240.0	$t_{2,2,2}^C$	440.01120358475225		
$dt_{hu,2}$	180.0	$b_{1,1,2}^{H,in}$	457.30751582696104		
$f_{1,1,2}^H$	0.7858259450291144	$b_{1,2,1}^{H,in}$	4913.424091100082		
$f_{1,2,1}^H$	7.5591139863078185	$b_{2,1,1}^{H,in}$	11800.0		
$f_{2,1,1}^H$	20.0	$b_{2,2,1}^{H,in}$	0.0		
$f_{2,2,1}^H$	0.0	$b_{2,2,2}^{H,in}$	4073.637821594076		
$f_{2,2,2}^H$	6.904470884057755	$b_{1,1,2}^{H,out}$	457.30751582696104		
$f_{1,1,2}^C$	3.107473915293866	$b_{1,2,1}^{H,out}$	4913.424091100082		
$f_{1,2,1}^C$	13.0	$b_{2,1,1}^{H,out}$	11800.0		
$f_{2,1,1}^C$	4.514661992235036	$b_{2,2,1}^{H,out}$	0.0		